Predicting New York City Airbnb Prices

Linear Regression Project – Summary Report

Objective

To build a model that estimates the nightly price of an Airbnb listing in New York City using its borough, room type, and review activity.

Data and Method

- About 48,900 listings from the Kaggle “New York City Airbnb Open Data” (2019).
- Cleaned the data: removed a blank row, dropped columns that do not help prediction (IDs, names, last review date), and filled missing review rates with 0 because those listings had no reviews.
- Removed extreme listings (above the 95th percentile of price and minimum stay) so the model focuses on the typical traveller market.
- Converted borough and room type into numbers (one-hot encoding).
- Trained a Linear Regression model on 80% of the data and tested it on the unseen 20%.

Results in Plain English

Metric	Value	What it means
R^2 Score	0.451	The model explains about **45%** of the differences in price between listings. The other 55% comes from things it cannot see, such as the view, interior quality, exact street, and season.
MAE	$37.45	On average, the model's price estimate is **off by about $37 per night**.
RMSE	$50.92	Higher than the MAE, which shows the model makes a few larger mistakes, mostly on expensive listings.

Where the Model Works Best

The model is more accurate for budget and mid-range listings than for expensive ones. Most listings are in the lower price range, so the model has more examples to learn from. High prices depend on luxury features (views, amenities, building quality) that are not in the data, and a straight-line model fits the high end poorly. The residual plot supports this: errors grow as the predicted price rises.

Key Findings

- Manhattan listings are priced higher than Bronx listings, even after accounting for the other factors.
- Room type has a strong effect on price: entire homes cost more than private or shared rooms.

Limitations and Recommendation

This model gives a useful **rough price guide** for hosts and travellers, but not an exact price. To improve it, add amenities, property photos, and exact neighbourhood, or try a stronger model such as Random Forest.